

opportunities for counterfeits. Although it is not clear how broad a role metrology would play, verifying the accuracy of specific design elements implemented to enhance hardware security could be one. An example is using the tuning, nonlinear conductance, reconfigurability, and variability properties of memristors[59] to create digital keys and fingerprints, where the encryption keys are derived from physical variations created during device fabrication[60].

Wafer manufacturers, process tool suppliers, pilot lines, and factory start-ups all have different timing and measurement requirements. The need for a shorter ramp-up time for pilot lines means that the characterization of tools and processes prior to pilot line startup must improve. However, as the process matures, the need for metrology should decrease. As device dimensions shrink, the challenge for physical metrology will be to keep pace with in-line electrical testing that provides critical electrical performance data.

2. SCOPE OF REPORT

The scope includes metrology for critical dimensions (CDs), microscopy for 3D and high aspect ratio (HAR) pattern measurements, overlay measurements, lithography metrology, front-end processes metrology, on-chip interconnect metrology, inter- and intra-die interconnect metrology, materials characterization, metrology for emerging materials and devices, reference materials, and outside systems connectivity needs and challenges. For the sake of simplicity, in the rest of this report we will group metrology needs for (a.) computation blocks and related new materials (i.e., FEOL), and (b.) connecting blocks (i.e., back- and middle-end-of-line), chiplets, back-side power delivery (BPD), and advanced packaging. In this differentiation we acknowledge that multiple intermediate blocks such as emerging memory and embedded storage can exist at the interface of these two blocks, with complementary needs.

Conferences and associated proceedings are important sources for researchers in different areas of metrology for nanoelectronics and IC manufacturing. As shown by the scope of the chapter, the breadth and depth of information and expertise needed to adequately address IC metrology issues are quite large. As such, some of the underlying information on instrumentation, characterization, device structure and physics, and process steps may not always make it into the chapter or in some cases are not suitable for a roadmap publication. Publications and specialized conferences that focus on IC metrology provide some of this background information, e.g., *Metrology and Diagnostic Techniques for Nanoelectronics* (Zhiyong Ma and David G. Seiler eds.) and *Handbook of Silicon Semiconductor Metrology* (Alain C. Diebold) [61] cover recent advances in research and industry practices on IC and nanoelectronics metrology, and are a good reference for a wide range of topics covered by the chapter. The Frontiers of Characterization and Metrology for Nanoelectronics (FCMN), the SPIE Advanced Lithography conferences, the International Symposiums for Testing and Failure Analysis (ISTFA), The Lithography Workshop, the IEEE International Electron Devices Meeting, the Electron, Ion, and Photon Beam Technology and Nanofabrication Conference, Advanced Semiconductor Manufacturing Conferences, and the VLSI Technology Symposium among others, are key venues for new research and development on nanoelectronics and nanoscale manufacturing/fabrication metrology. Papers and presentations from the FCMN (and predecessor conferences) covering more than 15 years are available[62]. Other sources of information include SPIE's *Journal of Micro/Nanopatterning, Materials, and Metrology* (*JM³*), *Nature Electronics*, *Advanced Materials*, *IEEE Transactions on Semiconductor Manufacturing*, *IEEE Transactions on Electron Devices*, and *IEEE Electron Device Letters*. The National Institute of Standards and Technology (NIST) recently published a series of workshop reports on the needs of the industry that highlight the technical and investment needs required to support a robust semiconductor manufacturing sector[63, 64]. The listed sources and chapter references are by no means exhaustive, but they are good places to start. In addition, the Metrology Focus Team (FT) occasionally publishes supplemental *white and technical papers* on semiconductor metrology topics to the roadmap[65, 66], [67],

3. SUMMARY AND KEY POINTS

This chapter presents the long- and near-term difficult metrology challenges. These challenges revolve around the impact of new drivers, device structure complexity, new materials, and the statistical limits of controlling sub-5-nm stochastic processes. A short section on metrology requirements for outside systems connectivity (OSC), is included. Also included is a section on virtual metrology requirements, its role, a summary of issues, and what needs to be done for wider adoption. Reoccurring themes throughout the report include a) the influence of advanced data analytics; b) the need for correlative and hybrid metrology, and c) the impact of new device architectures and new starting materials. These issues, which have been present in the roadmap over the years, take on new urgency as market drivers shift as outlined in the key messages below:

- Device structure and new materials complexities are as important as feature size as key drivers of metrology needs and challenges.
- Hybrid, data argumentation, and virtual / correlative metrology will play much bigger roles moving forward. No single instrument has the capability or resolution needed to measure the large number of parameters and measurands needed to characterize some of the new device structures.
- The use of advanced data analytics will span all areas of metrology including image and data analysis, inference, integration. Digital twins, modeling and simulation will drive prediction, process co-optimization, yield enhancement, cost and performance optimization. Although this already happens in some form or the other, current methods, which involve autonomous or semi-autonomous evaluation of data to learn patterns, obtain key insights, and make decisions, could fundamentally change how metrology is currently performed in the factory.

4. CHALLENGES

Rapidly changing technology needs, advances in lithographic and multi-patterning processes, new beyond CMOS materials, new device architectures, and switching devices drive metrology requirements. The rise of 3D NAND technology has greatly stretched the capability of current metrology. Multilayered stacked structures of 3D NAND (with hundreds of layers) present unique challenges with respect to vertical gate control, CD channel/memory hole, defect control, stress, and material characterization, among others. New channel materials, including SiGe/Si and III-V film stacks, higher dielectric constant materials, dual work function metal gates, wide bandgap semiconductors, and new ultra-shallow junction doping processes. 3D device structures, such as Nanosheet FET, CFET, 3D NAND, and 3D dynamic random-access memory (3D DRAM), place significantly more difficult requirements on dimensional and doping metrology. For example, fins are evolving into nanowires, nanosheets[68, 69], nanoribbons, or nanorods at future nodes[70]. 3D interconnect metrology requirements are largely driven by the activity in interposers (e.g., through silicon vias (TSV), and hybrid direct metal-to-metal bonding), back-side power delivery (BSPD) R&D. Several solutions for bonded wafer overlay are now available. For beyond CMOS R&D, many areas of 2D materials metrology have advanced [10, 65], but putting them into volume manufacturing will require additional challenging R&D. For example, the need for understanding large-area graphene uniformity is driving both physical and electrical metrology.

Extreme-UV lithography (EUVL) is driving the development of new metrology equipment for masks, and research into metrology techniques for EUV resist. EUVL-induced stochastic variations and defects require improvements in the sensitivity and resolution of metrology tools, and in models that relate EUVL patterning parameters to stochastic variations. Morden lithographic techniques require independent resolution of a large set of metrics to fully characterize a multi-patterning process, and depend on modeling. [71]. For example, such models assist metrologists in developing suitable measurement methods and strategies for EUVL-induced defects and variations.

4.1. NEAR-TERM AND LONG-TERM CHALLENGES

Metrology needs are currently dominated by the natural stochastic variation limits of the printing process, and by new structures, materials, and processes. Shrinking feature sizes, tighter control of device electrical parameters (e.g., tighter threshold voltage and leakage current) and new interconnect technology will provide the main challenges for physical metrology methods. Hence, to achieve desired device scaling, metrology tools must be capable of measuring properties on and at atomic distances.

Table MET-1 presents ten major challenges for IC metrology; these include in-line metrology for dimensional, compositional, and doping measurements. It is expected that many short-term (2024 to 2030) metrology challenges listed in Table MET-1 will continue beyond the 5 nm half pitch (HP) for physical gate length for HP Logic (i.e., well past 2039).

Table MET-1 *Metrology Difficult Challenges*

<i>Near-Term Challenges: 2024 -2029</i>	<i>Summary of Issues</i>
<i>Control of new process technology and 3D interconnects.</i>	• 3D device structures, such as nanosheet and Complementary-Field Effect Transistor (CFET), require an increased need for in-line metrology for dimensional, compositional, and doping measurements at the atomistic scale. Multi-patterning techniques introduce the need to independently solve a large set of metrics to fully characterize a multi-patterning process. • Multilayered stacked structures such as 3D NAND present unique challenges with respect to vertical gate control, CD channel/memory hole, defect control, stress, material characterization, and the high aspect ratio nature of the stacked memory cells, among others.
Statistical limits of sub-5 nm process control	• Controlling processes where the natural stochastic variation limits metrology will be difficult. Printed features are now small enough that random variations in the amount and position of molecules can now create small variations in pattern fidelity that have substantial effects on device performance.
Measurement of complex material stacks and interfacial properties.	• Capability for SOI, III-V, GeOI wafers needs enhancement. CD and film thickness are impacted by thin SOI optical properties and charging by electron and ion beams. • Reference materials and standard measurement methodology for complex material stacks and interfacial properties including physical and electrical properties are needed. • Optical measurement of gate and capacitor dielectric involves averaging over too large an area and needs to characterize interfacial layers.
Measurement of complex material stacks and interfacial properties.	• Overlay and other test structures are sensitive to process variation, and test structure design must be improved to ensure a correlation between measurements in the scribe line and on-chip properties. Standards institutions need rapid access to state-of-the-art development and manufacturing capability to fabricate relevant reference materials.
Metrology needs for 3D multi-chip integration (3D chip stacking)	• Metrology is needed for 3D multi-chip (memory, chiplet, or tile) integration technologies for higher density scaling of transistors using techniques such as TSV, embedded silicon bridge, 3D chip stacking using multiple-sized micro solder bump, and hybrid bonding. • Some applications need interconnect access from the wafer backside, which needs TSVs to connect devices and power supply sources outside the chips. TGV can be an issue but the glass trend is needed as PCB material replacement is confirmed. The backside power distribution network (BS-PDN) process requires new concepts of metrology over FEOL, BEOL, and A&P.

Long-Term Challenges: 2030 - 2037	Summary of Issues
Nondestructive wafer and mask-level metrology for 3D structures.	- Nondestructive, production-worthy wafer and mask-level metrology for CD measurement for 3D structures, overlay, defect detection, and analysis. - Potential metrology solutions need to be able to characterize complex shape structures with sub-atomic resolution (without surface charging and contamination). These include stacking nanosheet FET and CFET
Lab to Fab Metrology Strategy	Correlation of test structure variations with in-die properties is becoming more difficult as devices shrink. Sampling plan optimization is key to solving these issues. Techniques must provide mapping across the chip and across wafer variation.
Metrology for sub-3 nm multiple patterning EUVL, high-NA EUVL	Controlling processes where the natural stochastic variation limits metrology will be difficult. Printed features are now small enough that random variations in the amount and position of molecules can now create small variations in pattern fidelity that have substantial effects on device performance. Examples include EUV resist-induced surface, sidewall, and edge roughness, and stochastic effects.
Structural and elemental analysis	Materials characterization and metrology methods are needed to control interfaces, dopant placement, defect density, and elemental concentration relative to device dimensions. Measurements of self-assembling processes are also required.
Metrology needs for 3D device integration and 2D channel materials	3D integration technologies for transistor-level 3D structures are in the R&D phase toward mature process integration. Metrology for the characterization of subsurface structures (dimensional and compositional measurements) is needed. - Metrology for integration of proposed non-computing blocks structures (e.g., FeFET, MRAM, etc.), and memristors for neuromorphic applications. Metrology for CFET (monolithic/sequential), and photonic chip integration. - Metrology for 2D channel material: atomic defect characterization and monitoring (vacancies, substitutions and ad-atoms, grain boundaries, dislocations, strain, local variation of the band structure, and contamination, among others), and potential/carrier distribution.

5. TECHNOLOGY REQUIREMENTS

5.1. SUMMARY

This section summarizes the technology requirements over the next 15 years, and by extension the types of device structures and materials that would need to be measured. Table MET-2 is a subset of the More Moore – Logic Core device technology roadmap for select years between 2024 and 2039. It shows the projections of when devices from different technologies will be produced and some of the requirements needed for implementation.

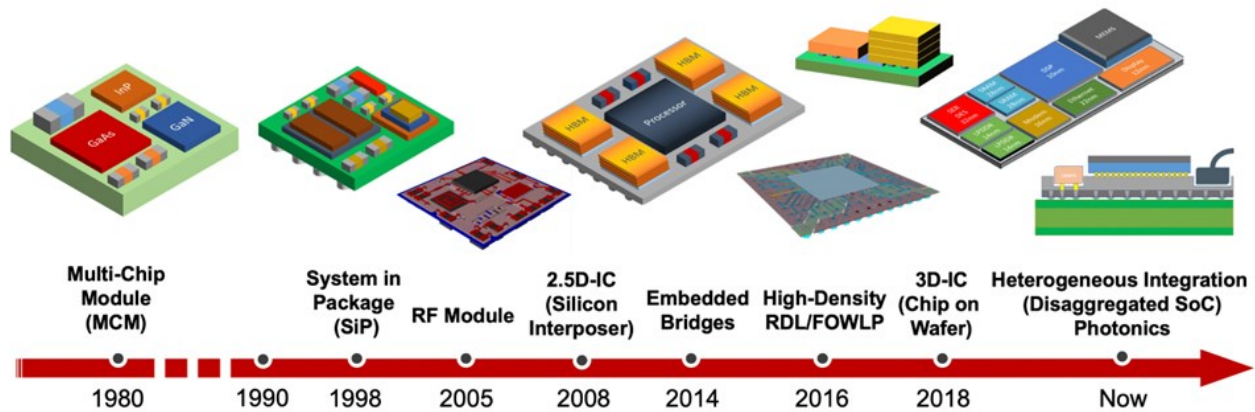


Figure MET7 Packaging Trends over the Years (1980 to 2024)

Metrology will play a key role in ensuring that these new design and manufacturing approach deliver the performance needed to address current and future computing challenges. Some of the metrology requirements for advanced packaging stem from the heterogeneity and density of chips being connected. Although chiplet is a design approach rather than the package, it will be a key driver for packaging innovations for years to come. Some of these challenges include dissimilar materials, complex thermal signatures, and different power requirements, more stringent alignment and placement accuracy, increased number of interfaces across different length scales, and chiplet testing methods electrical test binning are needed to solve known-good-die problem. Here, we highlight a few metrology needs. Some of the items below would be addressed in future roadmap sections.

5.12.1 Substrates

Packaging substrates connect the dies to the board and solves the interconnect mismatch between the output from the die(s) and input to the board and can contain embedded connections between dies. As such, advanced substrates that can support high-density interconnects are important enabling technologies for chiplet integration. In addition, packaging substrates could be rigid or flexible substrates, glass, semiconductor, or organic. This means that techniques or models developed for one application needs to be modified before they can be used for another.

Improved metrology solutions are needed for:

- Topography and Warpage
- Linewidth, spacing, and pitch.
- Film thickness.
- Coplanarity at die attach,
- Interlevel via diameter
- Thermal measurement for multilayer stacks.
- Electrical test on package

5.12.2 Hybrid Bonding

Although hybrid bonded chips are inputs to the package rather than the package, it is a technique for integrating multiple chips and delivers the highest density interconnect between chips. However, as a fab process, it also has some of the most stringent alignment and placement accuracy and planarity requirements compared with other packaging and assembly processes.

Key metrology requirements include:

- Planarity
- Surface roughness and topography and nanotopography at wafer, die level.
- High precision alignment, tilt control, and placement accuracy.

- Bonding interface defect and particle detection
- Organic molecular contamination
- Warpage / Bump size and defect characterization.
- Fine pitch probing methods
- Edge inspection
- Void Detection
- Best testing and binning methods - known good dies.
- Adhesion and Adherence energy in adequate condition (e.g. anhydrous)

5.12.3 Thermal Management

The integration of devices with different thermal profiles and materials with dissimilar coefficients of thermal expansion in a single package means that thermal management and by extension thermal metrology are critical requirements for advanced packaging. The high-density integration of chiplets can restrict heat spreading. Thermal metrology requirements include:

- Interface thermal characterization techniques
- Heat load management and validation
- New system-cooling technologies
- New thermal metrology characterization tool
- Thermal uncertainty models

5.12.4 Modeling and Simulation and Validation

Better modeling capabilities are needed for high-density interconnects, fine pitch scaling, new materials, more stringent alignment and placement accuracy, increased number of interfaces across different length scales, and complex thermal signatures. To ensure accuracy and reliability, these models would need to be validated using metrology and performance data. Modeling and simulation needs include:

- Complex multi-physics, multi-stage, and multi-scale models to predict the performance of the assembled package.
- AI and ML assisted design and simulation approaches for package design
- AI and ML assisted metrology and test

5.11. 3D NANOMETROLOGY NEEDS AND CHALLENGES

The simultaneous decrease in feature dimension and increase in 3D structural complexity of advanced device architectures is imposing daunting challenges to the semiconductor metrology community. In fact, the need for better electrostatic control has led to fully depleted silicon-on-insulator (FDSOI) devices as well as the replacement of relatively planar structures with devices with much greater topological complexity such as FinFETs, tri-gates, gate-all-around structures or even stacked gate all-around-structures[293, 294]. For instance, evaluation of structural parameters of 3D shapes requires measurement of many parameters beyond traditional CD, including sidewall angle, top corner rounding and footing, roughness, recesses, and undercuts. Now all are indispensable, as are thin film parameters such as thicknesses on horizontal and vertical surfaces, compositions, dopant distributions, strains, and other parameters and materials properties within device structures. Future 3D memory devices will also include multiple gate-level structures defined by HAR trenches and holes in multilayer stacks with as few as 4 alternating gate/dielectric periods, but possibly as many as 256. The lack of suitable metrology of HAR features is currently a major gap, as in-line non-destructive methods do not have the needed sensitivity and resolution to image or measure CD, depth, profile, or contamination at or near feature bottoms[112]. Furthermore, these problems are compounded by the very small dimensions of the devices, where only a few atoms yield extremely small signals, and by limited sample resilience, random distributions of dopant atoms, non-uniform nanometer-scale strain fields, and quantum confinement effects. Thus, 3D characterization at sub-nanometer and maybe atomic resolution levels is increasingly required to understand the physics of devices controlled by surface and interface effects, and such characterization will become critical for process development if not high-volume manufacturing.

Today, advanced devices are characterized at the nanometer level, with mainly offline, TEM-based techniques such as NBD, dark field holography, EELS, etc. However, the intrinsic 2D cross-section information (or projection) of all these techniques is not complete since the assumption of conservative information through the third dimension is no

longer applicable. This makes it difficult to use 2D information to extend to the 3D parameterization of the devices. Thus, some new approaches like tomography reconstruction will need to be implemented. All the above-mentioned techniques bring about issues with sample preparation and characterization yield due to the increased number of off-line characterizations and with the link and balance between in-lab and in-fab metrology.

TSOM is inherently a 3D metrology method. It has been applied to study 3D shape of sub 40 nm features. From simulations and limited measurements, the TSOM method shows promise for 3D shape metrology but needs further study beyond simulations for applications in production[254].

In the following section, an overview of the state-of-the-art promising techniques for nanometrology is presented, along with their potential for tomography 3D information reconstruction. Also, the need to combine the different 3D information, linked to die-level and wafer-level maps for HVM metrology control, will be discussed.

5.11.1. 3D NANOSCALE CHARACTERIZATION OF SINGLE DEVICES

There are basically three approaches to mitigate the 3D complexity: true 3D imaging techniques, 3D reconstructions from 2D images, and simplified test structures designed to enable the extraction of 3D information from several 2D measurements.

5.11.1.1. STRUCTURAL: (DIMENSIONAL, CRYSTAL QUALITY)

To get access to structural properties at the device level, TEM-based techniques have the advantage of easy positioning for local analysis. TEM is widely used to accurately measure device dimensions due to its excellent, sub-0.05 nanometer resolution while scanning electron microscopy (SEM) has a resolution limited to approximately 0.5 nm. Focused ion beam (FIB) technology allows the preparation of thin lamella or tip-shaped samples of a specific device from a cleaved/diced or a full wafer[295]. With these two types of sample preparation, it is possible to use high-resolution (S)TEM imaging for 2D structural analysis at the atomic scale, or electron tomography to measure the dimensions of the 3D architecture of the advanced nodes. Moreover, with the recent development of parallel nanobeam diffraction (NBD) and electron precession, mapping of crystal orientation (e.g., copper interconnection lines) is possible with an increased spatial resolution (around 3 nm). Even if there is now an industrial solution with FIB systems to be able to automatically prepare and extract samples for TEM imaging, and use dedicated imaging software, usually the technique is time-consuming and site-specific, with lamella that are 100 nm thick and a few μm long[296].

FIB/SEM 3D refers to the acquisition of high-resolution scanning electron microscopy (HRSEM) images combined with FIB slicing. Today's state-of-the-art has been shown with a volume of an array of 22 nm node transistors. Despite its potential, some problems still need to be solved. The resolution of the reconstructed volume depends strongly, but not only, on the SEM resolution. In addition, such metrology is very time-consuming and might present a problem of low success rate due to sample drift and tool instability over a long period of time. Software solutions have already hit the market to partially overcome this problem, but for a reliable in-line 3D metrology with SEM, dedicated equipment still needs to be designed.

Atom probe tomography (APT) is particularly well suited for measuring composition at the atomic scale in semiconductor devices. The ability to detect all elements with detection limits as low as 10^{17} atoms/cm³, for B in Si host makes it an attractive technique. The sample must be prepared as a tip shape, e.g., by FIB milling, to enable laser-assisted field evaporation. Placing a complex transistor structure at the end of an atom probe tip may be challenging, but in most cases is possible by a skilled FIB user. The tip shape and field evaporation impose an analyzed volume of, typically, some 80 nm $\times$ 80 nm $\times$ 200 nm, which although small, matches well with current and future device dimensions. Although reconstruction image distortion is a key error source, the use of *in situ* TEM or SPM could help reduce such errors[297].

Analytical TEM allows identification, and in some cases, quantification of elements to an atomic resolution with C_s aberration correctors[298] but with a lower sensitivity than APT or secondary ion mass spectrometry (SIMS)—around 1%. Electron energy loss spectroscopy is traditionally well suited to analyzing lighter elements, with energy-dispersive EDS for heavier elements, but EELS is the only technique able to differentiate the chemical bonding of probed elements. Recently the speed of spectral acquisition has been increased with the development of silicon drift detectors (SSD) for EDS, and the time to acquire elements or bonding states has been dramatically reduced (by a factor of 10 or more) with fast electronics for EELS imaging filters also. Thus, as far as analytical electron tomography is concerned, the results now are encouraging[299].

5.11.1.2. PHYSICAL: (STRAIN, STRESS, ELECTRICAL)

True 3D imaging techniques include coherent X-ray diffraction and small-angle X-ray scattering. For coherent X-ray diffraction, the use of coherent X-rays simplifies the phase problem of inverse Fourier transforms and permits the extraction of 3D scattering features from the recorded intensity of their diffraction patterns. SAXS in transmission at multiple angles can provide full 3D information but provides the average of feature geometries over the X-ray spot size. However, as of today, only synchrotron facilities can provide this kind of analysis[300, 301]. (*See the Lithography Metrology section for a description of CD-SAXS*)

TEM techniques for 2D strain measurements are now available, depending on the instrument's performance and stability and on the area to analyze: dark-field holography (3–10 nm resolution and a sensitivity of $\pm 0.06\%$)[302], Nanobeam electron diffraction (NBED) (3–6 nm resolution and an excellent sensitivity of $\pm 0.02\%$), high-resolution TEM and now STEM high angle annular dark field (HAADF) with geometrical phase analysis (1 to 5 nm resolution and sensitivity of $\pm 0.1\%$)[303] and convergent beam electron diffraction (1 to 5 nm resolution and excellent sensitivity of $\pm 0.02\%$). Today, NBED in precession mode (N-PED) seems to provide the best combination of spatial resolution, sensitivity, and robustness from artifacts. In addition, it is very insensitive to thickness variations in the specimen, and thus time-consuming preparation techniques such as backside milling are not required. The experiment and processing can take between a few minutes to several hours depending on the size of the map acquired. This will speed up in the future by improvements in imaging cameras and real-time data processing. Currently, NPED is not widely available in electron microscopes using the imaging capabilities of the CCD camera. However, it is possible to use a commercially available off-axis television camera to measure the positions of the diffraction spots for deformation mapping and a sensitivity of 0.03% and 4 nm spatial resolution has been reported on a pre-aberration correction generation microscope[304]. The challenge is to have well-controlled sample preparation to achieve good reproducibility of measurements[305]. In addition, more precise quantification of the metrology performances (accuracy and precision) should be discussed[306,307]. (*See the section on Strained Si Processes*).

Electrical tomography using scanning spreading resistance microscopy (SSRM) has been recently developed[231, 308]. It is an emerging solution to address 3D carrier mapping. The methodology consists of using an AFM probe to remove material by successive scanning and to record at every scan the 2D carrier map, similar to a slice-and-view methodology. The resulting stacked images are interpolated and are used to generate a 3D. This method requires a hard-conductive probe (such as a full-diamond tip) to ensure material removal without significant damage to the probe. This approach has been highlighted in carbon nanotube-based interconnects and various emerging memories. By increasing the accuracy in the tip-induced material removal, sub-nm removal rates have been obtained leading to the characterization of volumes in the order of 100 nm^3 [309]. This approach is referred to as scalpel SPM and has been used for the experimental observation of conductive filaments in integrated resistive switching devices, thus enabling the 3D characterization of confined volumes in oxides such as Al_2O_3 and HfO_2 with removal rates between 0.2–0.5 nm/scan[310, 311]. The fast Fourier transform-SSRM (FFT-SSRM) appears also as an emerging method to image highly doped samples[229]. By modulating the load force applied to the tip, only the contribution of the spreading resistance is measured[229]. This is extracted by an FFT-based lock-in deconvolution method. This approach has been recently illustrated for a FinFET structure with a highly doped (1–4 m $\Omega\cdot\text{cm}$) raised source/drain region. In this example, FFT-SSRM allows to detect the highly doped SiGe and the undoped Ge regions without saturation induced by parasitic resistance as obtained with the standard SSRM technique[230, 312].

5.11.2. TRENDS IN NANOSCALE TOMOGRAPHY

All the above-mentioned techniques have been developed at different length scales for morphological, chemical, electrical, and structural imaging, and their 3D characterization capabilities are largely based on tomography. Important tomographic techniques for 3D imaging at microscopic and nanoscopic length scales are:

- X-ray tomography:
- FIB/SEM 3-D
- TEM/AFM
- Electron tomography
- Atom probe tomography

In any case, sample preparation is crucial; experimental techniques and sample preparation methods for tomography are both reviewed elsewhere[313]. Hard X-ray nano-tomography at 20 nm resolution may now come of age, as it is an ideal technique because it has the potential of non-destructively providing high-resolution images of buried devices. However, because of physical constraints, no commercial system exists that simultaneously allows for high resolution

(100 nm or less), penetrating X-rays (15 keV or more), and high throughput, and the technique is currently confined to 3rd generation synchrotron sources. Commercial systems can either accommodate high resolution at low energy or medium resolution at high energy in the case of large samples[314]. The construction of dedicated nanoelectronics synchrotron end-stations for 3D imaging is currently the only solution to get routine and fast access to the technique[315].

Electron tomography using STEM-HAADF contrast is now a routine and mature technique operating at about 1 nm resolution for those samples having a cross-section of 100 nm or less[263, 316-318]. Atomic resolution has been proved[319] but on simple, isolated objects like nanoparticles. Obtaining atomic resolution on complex objects is a real upcoming challenge, which could be overcome by improving the overall stability of the microscopes, optimizing the depth-of-field, the sample preparation (which is crucial, yet time-consuming), and data pre- and post-processing, presumably with machine learning (ML) and artificial intelligence (AI).

FIB/3D SEM is a technique of choice: it allows for relatively high-resolution (20 nm–50 nm) investigation of large objects with different contrast modes simultaneously, in such a way that the 3D sample shape, chemical composition, and even crystalline distribution can be retrieved[320, 321]. Neither X-ray tomography nor electron tomography can provide all this at the same time at such a resolution. It requires limited sample preparation (no need to prepare needle-like samples) but is destructive: this is not a severe limiting issue in nanoelectronics except for those cases where imaging modalities need to be cross-correlated. However, the method is still time-consuming and can be limited by drift issues, but so far it offers the best compromise among price, information, time, resolution, and sample size. One could think of a cluster of dedicated dual-beam systems for automated 3D high-throughput analyses. For compositional analysis, such cluster systems could include both EDS and SIMS detectors.

Recently, TEM and AFM have also been combined for 3D imaging beyond topography. Using the 3D analysis capability of scalpel SPM and a dedicated TEM sample preparation, nm-resolved 3D electrical and chemical analysis of Si- and Ge-based structures for sub-20 nm technology nodes have been studied[322]. This method is destructive, and it requires a sample preparation comparable to FIB/SEM. However, for site-specific studies such as failure analysis, this approach can be extremely effective in offering a solution to sense multiple material properties by using the entire spectrum of different AFM modes including electrical and non-contact. In addition, microwave tomography, e.g., electro-capacitive tomography (ECT), may be useful in the visualization of subsurface defects; however, the data processing requires situation-specific algorithms and numerical models.

The 3D Atom Probe or Atom Probe Tomography (APT) is an advanced version of a field ionization microscope combined with a mass spectrometer capable of atom-by-atom three-dimensional reconstruction of a small needle-shaped sample. The sample may be prepared from a device-specific site by chemical or plasma etching, or focused ion beam lift-out techniques similar to those commonly used for TEM sample preparation. In the 3D Atom Probe experiment, a needle-shaped sample is placed near an electrode and a strong applied field ionizes atoms from the sample tip, ejecting atoms from the sample, and accelerating them through a position-sensitive mass spectrometer. The original position of atoms in the sample is determined from geometric considerations and the atomic mass is determined from the time of flight. 3D Atom Probe provides a means of measuring the atomic arrangement of free-standing wires allowing a method to characterize doping density in nanowires.

APT is the most spatially-resolved mapping technique, featuring 0.1–0.3 nm resolution [323-325]. While not as mature as some of the techniques discussed above, the results are obtained from extremely small volumes (100 nm³). However, the wide variety of materials now used in advanced devices complicates atom probe analysis because the evaporation field is different for each material and these differences lead to artifacts in the reconstructed volume, but progress has been made with the addition of laser pulsing.

Overall, APT brings us closer to the dream of atomic mapping in three dimensions. Current detection efficiency is approximately 80% of the atoms ionized and there has been much progress in developing an understanding of local field effects that affect resultant 3D models. It must be emphasized that it is not enough to know just the position, identity, and chemical nature of all atoms in a material; it must be accompanied by microstructural information to discern the materials' properties and behavior. Thus, synergy is expected between the experimental determination of atom positions, identities, and chemical nature and computational material science to obtain the optimum utility of APT[280].

These reconstruction artifacts limit the accuracy of APT composition profiles, and the data requires careful calibration by other techniques, such as TEM and electron tomography. As the sample preparation for APT is similar to electron tomography, one interesting possibility is to perform electron tomography and then APT on the same sample, allowing

the APT volume to be corrected using the morphology measured by electron tomography. These types of measurements can also be correlated with other measurements (SPM[235], SIMS, etc.), combining the advantages of measuring many devices at once with nm-scale compositional measurements in 3D. The uptake of atom probe tomography for semiconductor analysis is increasing and with future improvements in detectors, reconstruction algorithms and tip shape monitoring will become an increasingly essential tool.

The emergence of the related technique of FIB/ToF-SIMS tomography allows complementary (better detection of light elements, isotopic information) chemical information with respect to FIB/SEM-EDS to be acquired[326]. Both serial sectioning techniques can be combined with X-ray nano-tomography to assist in targeting the region of interest and in reducing artifacts.

The accommodation of the operando experiment is illusory so far, but ex-situ or even in-situ experiments may be feasible with X-ray probes. A generic technique does not exist, and the choice of the tomographic probe is largely sample-dependent; however, dual beam FIB-SEM serial sectioning gives a satisfying trade-off provided samples are not precious. For high-throughput 3D nanoscale characterization, the tools need to be more stable, more user-friendly, automated, more flexible, and versatile. It must also be emphasized that sample preparation is crucial for each of the discussed techniques.

Computer aided X-ray-based techniques, e.g. ptychographic X-ray computed tomography (PXCT), and ptychographic X-ray laminography, can now nondestructively reconstruct the interconnects in a chip layer by layer and in 3D. The tomography has been used to capture interesting details such as the imperfections in the BEOL copper interconnects between the layers, and the roughness between the copper and the interlayer dielectric. [YO7], [YO8]. Furthermore, ultrahigh resolution, nondestructive, and rapid imaging of marginal fault detection in large, intact packages advanced packaging has been demonstrated [327].

5.11.3. DATA ANALYTIC (AI/ ML/ DT) FOR AUGMENTED COMPLIMENTARY HYBRID METROLOGY

It seems inevitable to use more than one type of measurement, tool, and method, and take advantage of the rich, multi-faceted set of data by combining the most reliable information gained from each source. This combined (complementary, hybrid) metrology would be necessary for sub-10 nm dimensional metrology. Hybrid metrology has shown promising advantages in significantly reducing the overall uncertainties, i.e., augmenting the confidence in the results, and in the optimization of various complementary measurement methods for cost[159, 200, 328-332]. More specifically, 3D data acquired for dimensional metrology will need to be compared and merged to that coming from other parameters (e.g., chemical, strain, electrical). In this sense, correlative microscopy recently has shown the advantages of such an approach while characterizing nano transistors[333]. They will not necessarily have the same resolution, as they will come from different tools. Therefore, modeling data analysis and treatment algorithms to merge data will need to be tested to ensure against artifacts resulting from the stitching/merging process.

Both CD-SAXS and GI-SAXS could be used to improve optical scatterometry through combined or hybrid measurements. Often there are high correlations between model parameters used in optical scatterometry or parameters to which optical scatterometry is insensitive. Deeper layers or buried metal layers are often opaque to optical metrology techniques but can easily be measured by CD-SAXS or GI-SAXS. Given one or more parameters in common between each model geometry[29] a small amount of SAXS data such as multiple patterning errors can be incorporated into the optical models, improving the optical measurements and reducing parameter correlations, while maintaining a high throughput measurement. 3D-AFM data[334] is already used with both SEM and scatterometry[159, 200, 335] but given the intrinsic capabilities of probe microscopy, a 3D-AFM with material contrast would be a good hybrid tool. Certainly, some feedback with sample preparation will be needed to avoid or diminish such artifacts and accomplish the goal of complete 3D information describing the device properties. Additional efforts are needed in quantifying uncertainty and error analysis in combined metrology where data fusion from different instruments is non-trivial. As the technique gets widely adopted, a common evaluation guideline for uncertainty on combined metrology would be needed.

An application of combined metrology is the Reference Measurement System (RMS), where one of the instruments provides traceability to the others. An RMS could be a single instrument or a set of several instruments. An RMS is well characterized using the best the science and technology dimensional metrology can offer, applied physics, sound statistics, related standards, and proper handling of all measurement error contributions based on the best protocols available. Because an

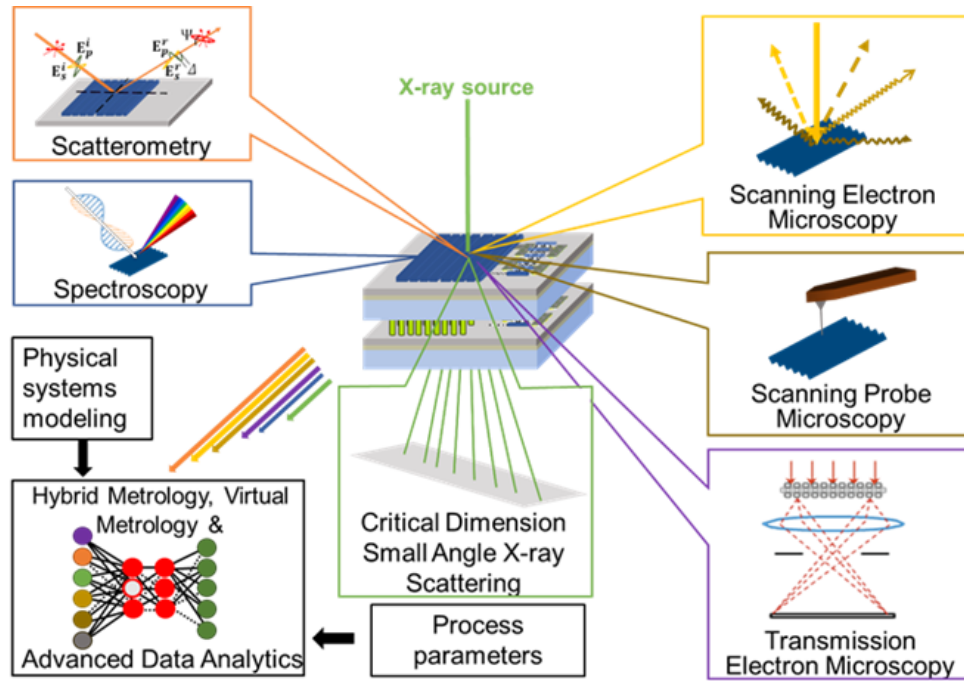


Figure MET8 A Multiple Techniques Approach to Device Metrology

RMS has been well characterized, it is more accurate, perhaps by an order of magnitude, and more precise than any instrument in a production fab[330]. An RMS must be sufficiently stable that other measurement systems can be related to it. An RMS can be used to track measurement discrepancies among the metrology instruments of a fab, control the performance and matching of production metrology instruments over time, and introduce absolute traceability to the measurement process if desired. Due to the performance and reliability expected from this instrument, the RMS requires a significantly higher degree of care, scrutiny, and testing than other fab instruments. Through its measurements, this “golden” instrument can help production and reduce costs. The main difference between RMS and combined metrology in general is the relationship between the instruments. In other applications of combined metrology, each instrument is better suited for a specific measurand but may not necessarily derive their traceability from the group. (See the section on Measurement Uncertainty.)

5.11.4. METROLOGY MATURIZATION (TRANSITION FROM LAB. TO IN-FAB)

For in-fab metrology, it is necessary to map nanometer-scale material properties over the entire wafer (e.g., optical properties, strain, critical composition), critical shape, and other critical functions. These properties are all measurable on conventional flat films, but the various properties in the device features often will not correlate well to those on traditional solid test pads; thus, many of these properties will need to be measured on complex 3D structures. When included within the more complex architectures that influence such properties, extracting these from among the many other unknown parameters will likely be difficult. As has been shown previously the only techniques available today for 3D nanometrology are more in-lab techniques. However, performing such metrology non-destructively is the ultimate challenge for cost-effective HVM[69].

Presently, the industry is using available traditional failure analysis laboratory techniques, but it is increasing the amount and speed of collected data through automation. The drawback of using such “brute force” methods is the obvious loss of wafers to further processing, which not only increases costs and cycle times for learning but also increases the uncertainty in the correlation between the measured properties and final device performances due to needs to compare results from different wafers. The industry must decide if this strategy is to be used in the long term; it might be feasible to build tools, which only damage a very small area on a wafer. This small region, after the measurements, can be neutralized to allow further processing. As indicated previously, such a tool would include milling, and several imaging and analysis techniques bundled together. It would be expensive but might offer a solution